



qbio
quantitative
biology

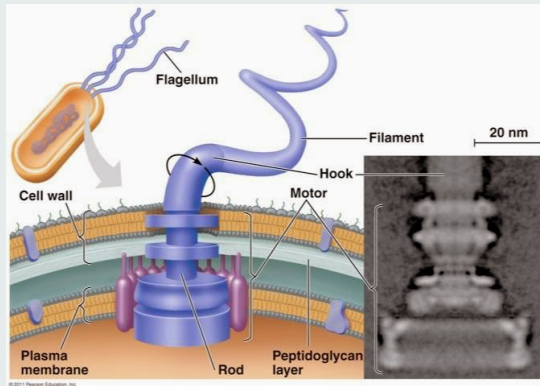
INTRODUCTION TO QUANTITATIVE BIOLOGY

Overview on the field

17TH SEPTEMBER 2026

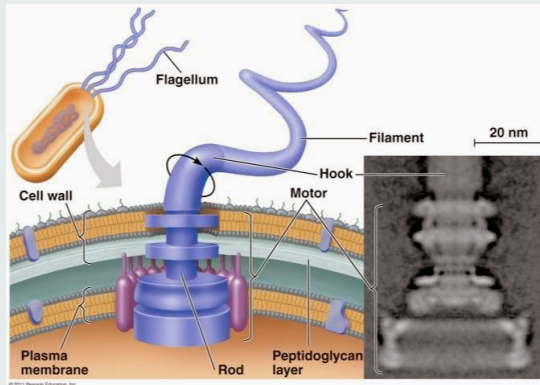
Luca Ciandrini (luca.ciandrini@umontpellier.fr)

Biological System



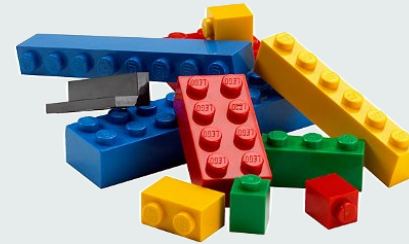
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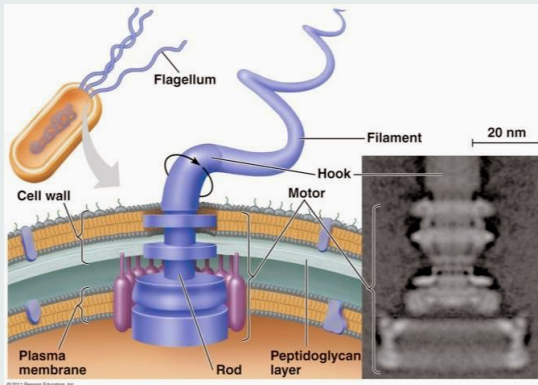
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Components



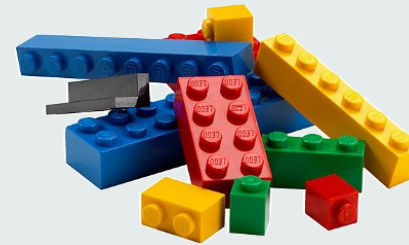
Characterisation of
biological parts
(Toolkits)

Biological System



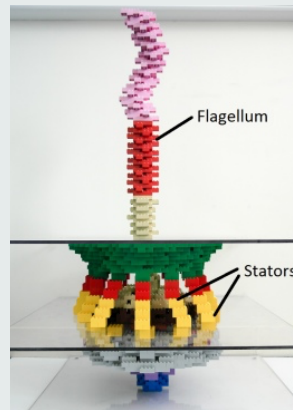
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Components



Characterisation of biological parts (Toolkits)

Systems Biology

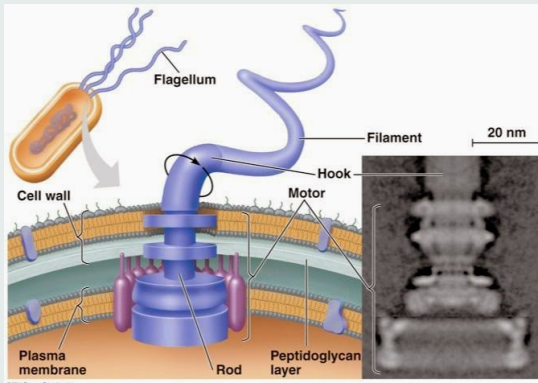


Imperial College of London

“Systems biology is the study of biological systems whose behaviour cannot be reduced to the linear sum of their parts’ functions. Systems biology does not necessarily involve large numbers of components or vast datasets, as in genomics or connectomics, but often requires quantitative modelling methods borrowed from physics.”

from nature.com

Biological System



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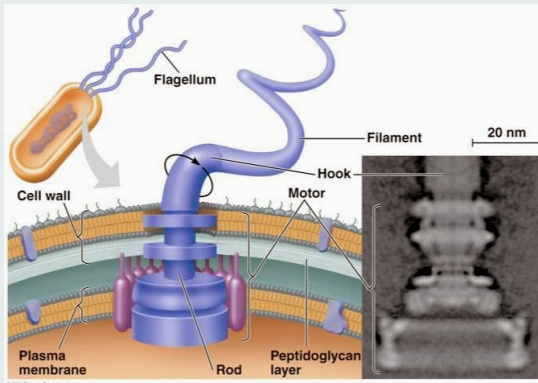
Synthetic Biology

“Synthetic biology is the design and construction of new biological parts, devices, and systems, and the re-design of existing, natural biological systems for useful purposes.”

from nature.com

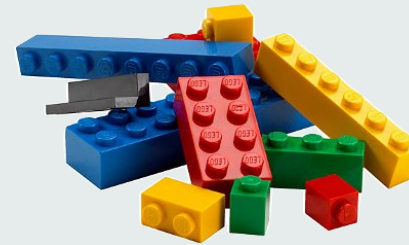


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numbers and tools to obtain predictions and biological understanding

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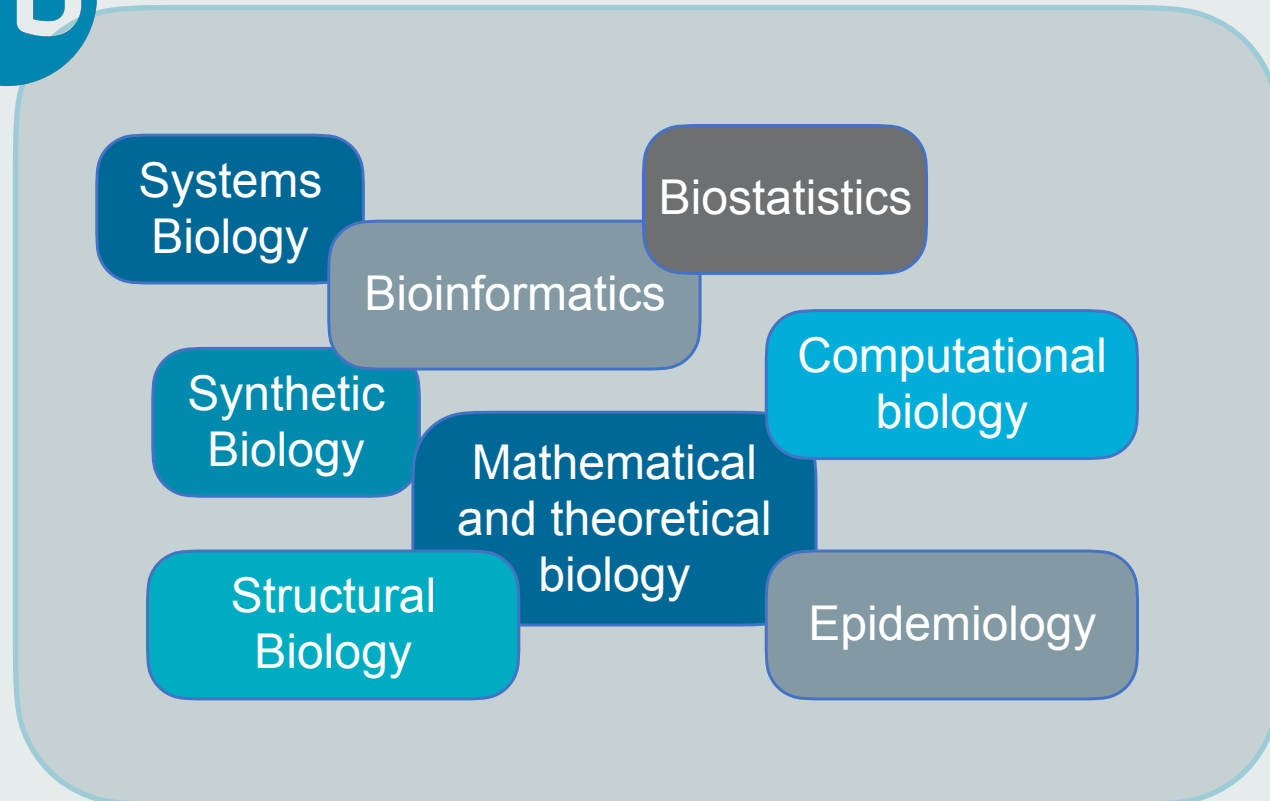
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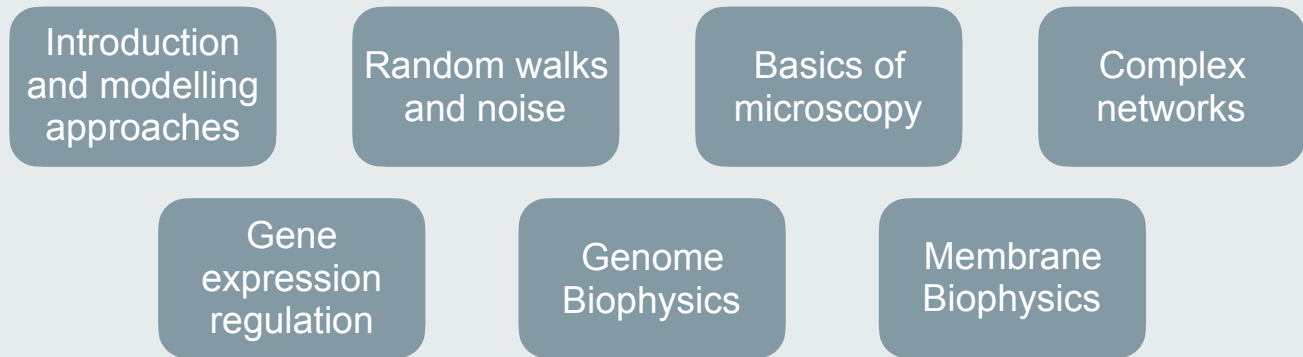
qbio $\neq$ biology + quantitative tools

numbers and tools to obtain predictions and biological understanding



Quantitative Biology





Timetable. Look at our GDoc.

We will give you some material (videos, pdf,...) to revise **before each lecture**.

Please check the GitHub of the course*:

<https://github.com/qbiomaster/qBioMaster-introduction>

IMPORTANT: learn how to manage your time!

**inform us of any incoherence with the GDoc*

Assignments

We mainly *work on projects*, with short reports that you will have to return and that we will grade.

5 assignments x 20%. More in details

- 2 organised by Luca (Modelling, Gene expression,...)
- 2 organised by Marcelo (Genome biophysics, microscopy,...)
- 1 organised by Pierre-Emmanuel (membrane biophysics)

We have different formats (ask the person!)

The numbers of a cell

\ Reminder: the central dogma

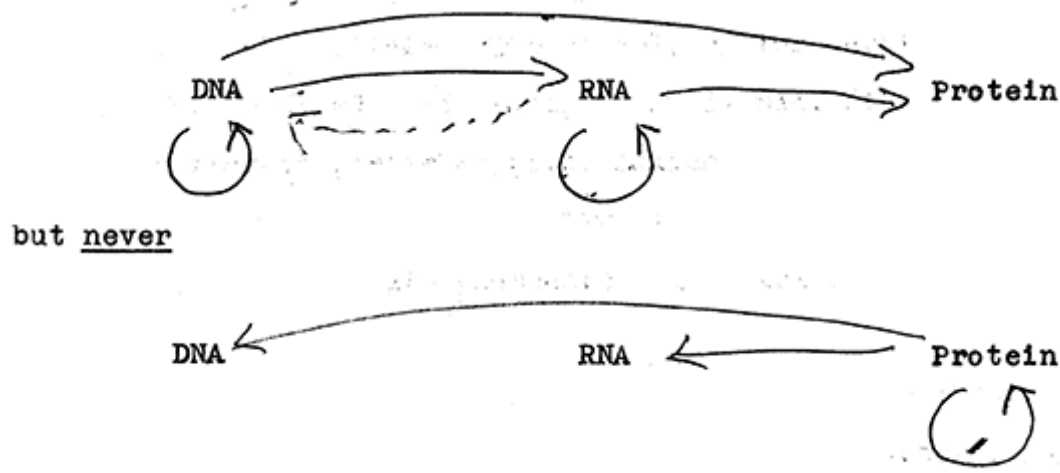
\ Let's give some numbers

\ Focus on timescales of gene expression

1. The numbers of a cell

\ Reminder: the central dogma

The Central Dogma: "Once information has got into a protein it can't get out again". Information here means the sequence of the amino acid residues, or other sequences related to it. That is, we may be able to have



where the arrows show the transfer of information.

Francis Crick, 1956
never published.

Then Crick wrote
The central dogma of Molecular Biology
Nature, 1970 [[link](#)]

1. The numbers of a cell

\ Let's give some numbers

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\ Let's give some numbers

question (<i>E. coli</i>)	Alon, Table 1.1
proteins per cell	4×10^6
genes	4500 $\Rightarrow$ ~ 900 copies per protein type on average
time to translate a protein	$\lesssim 1$ min (20 aa/s)
typical protein lifetime	1 h
cell generation time	20 min to several hours
1 protein per cell	≈ 1 nM

From Table 1.1, *An introduction to systems biology*, U. Alon

1. The numbers of a cell

\ Focus on timescales of gene expression

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\ Focus on timescales of gene expression

TABLE 1.2 Typical Timescales for the Reactions in the Transcription Network of *E. coli*

Binding of a small molecule signal to a transcription factor, causing a change in its activity	~1 msec
Binding of an active transcription factor to its DNA site	~1 sec
Transcription and translation of the gene	~minutes
Timescale for 50% change in concentration of the translated protein (stable proteins)	~1 hour (1 cell generation)

An introduction to systems biology, U. Alon

THE SIMPLEST MODEL OF GENE EXPRESSION

What is gene expression?

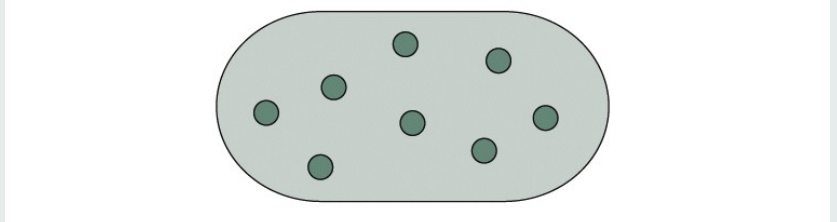
THE SIMPLEST MODEL OF GENE EXPRESSION

Let's build a model together [constitutive expression]

NOT EVERYTHING CAN BE MODELLED WITH ODE...



Ex: Cell-to-cell variability



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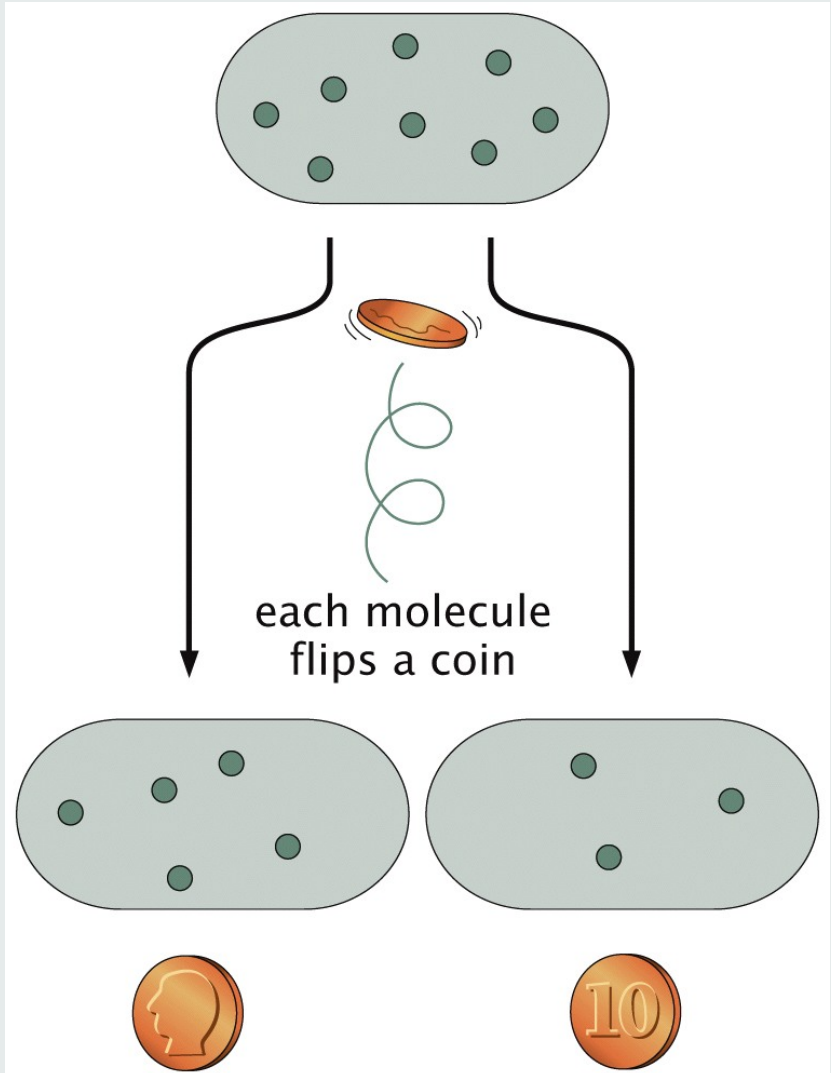


Figure 2.7a Physical Biology of the Cell, 2ed. (© Garland Science 2013)

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Ex: Cell-to-cell variability

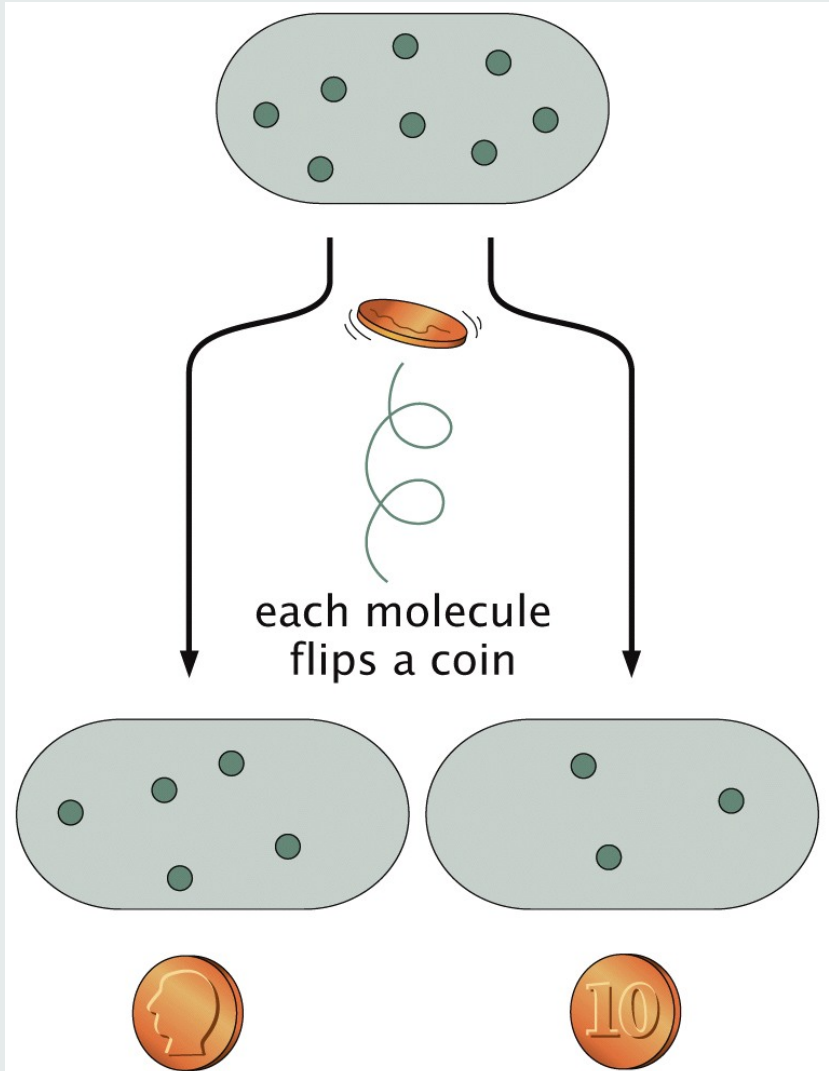
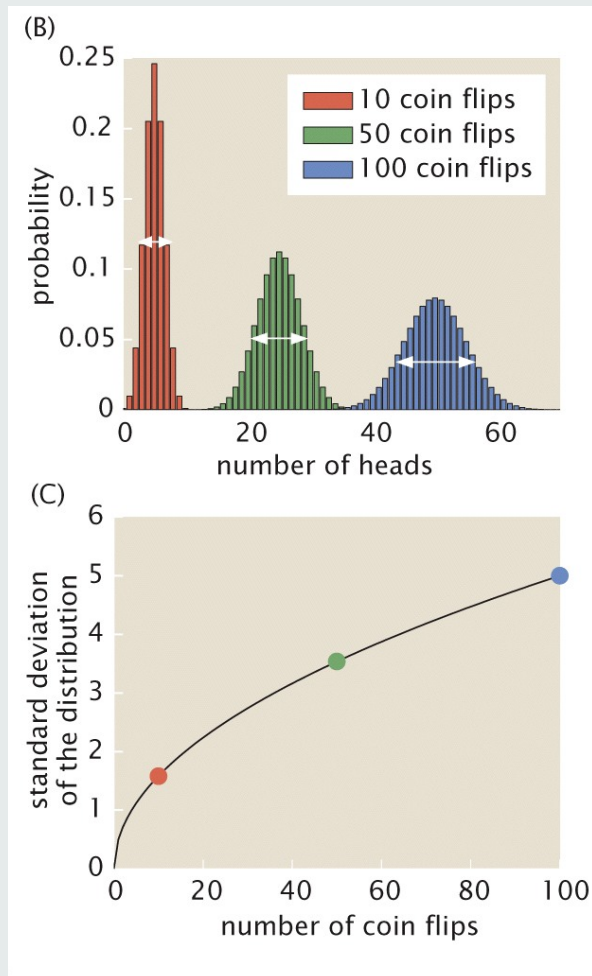


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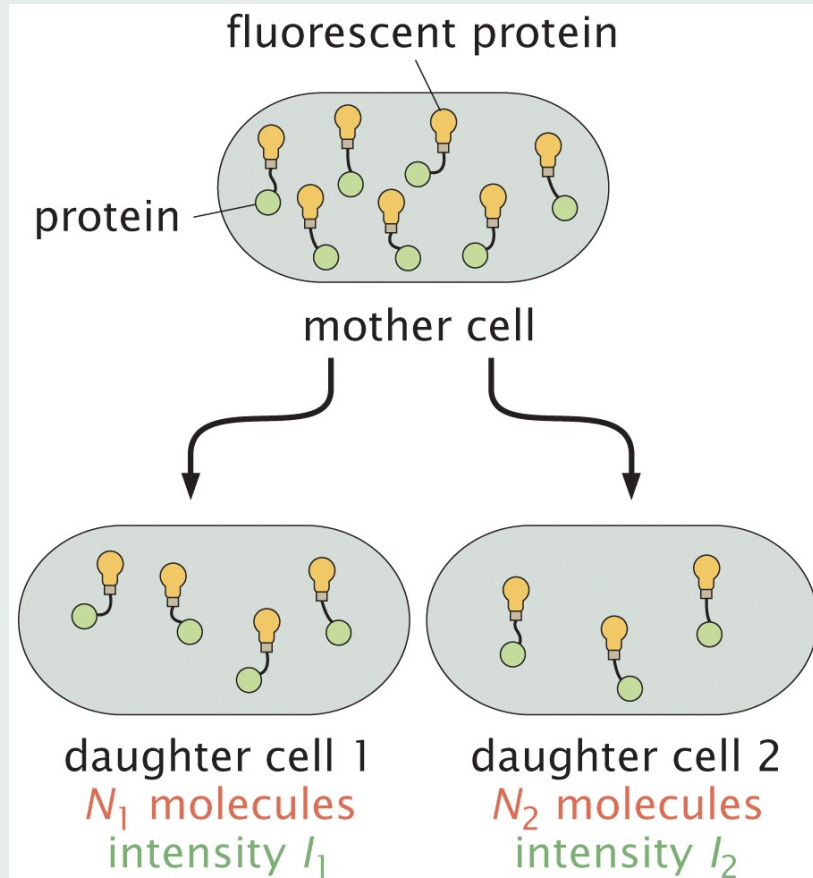


Figure 2.10a Physical Biology of the Cell, 2ed. (© Garland Science 2013)

We can derive that:

$$\langle (I_1 - I_2)^2 \rangle = \alpha I_{tot}$$

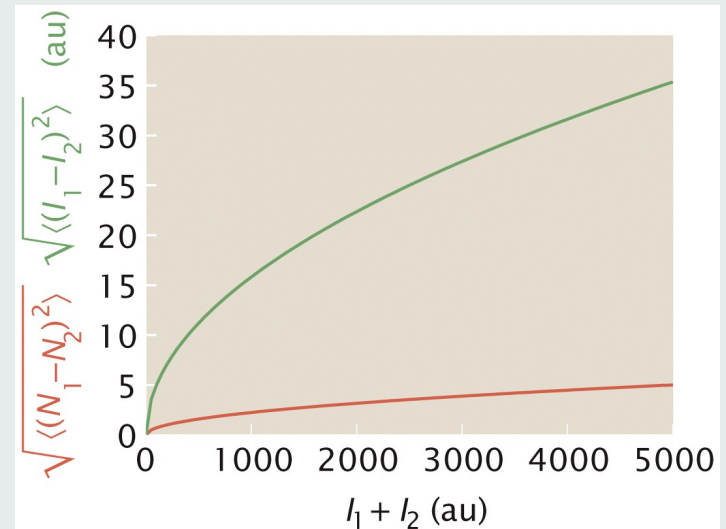


Figure 2.10b Physical Biology of the Cell, 2ed. (© Garland Science 2013)

Data from Rosenfield et al., *Science* 307:1962 (2005)

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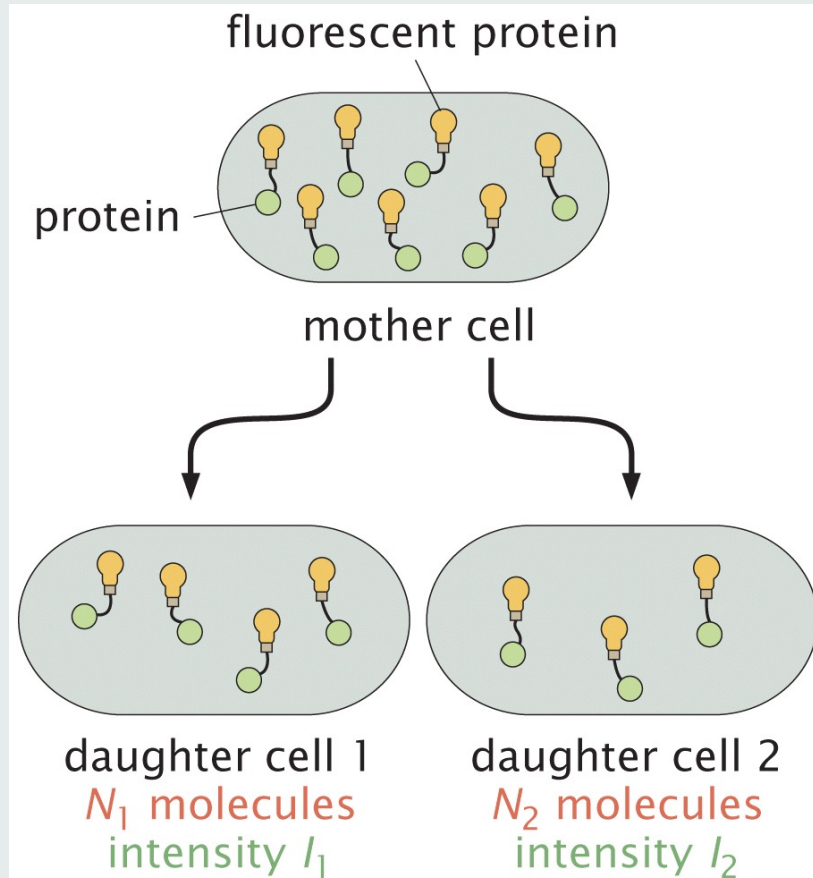


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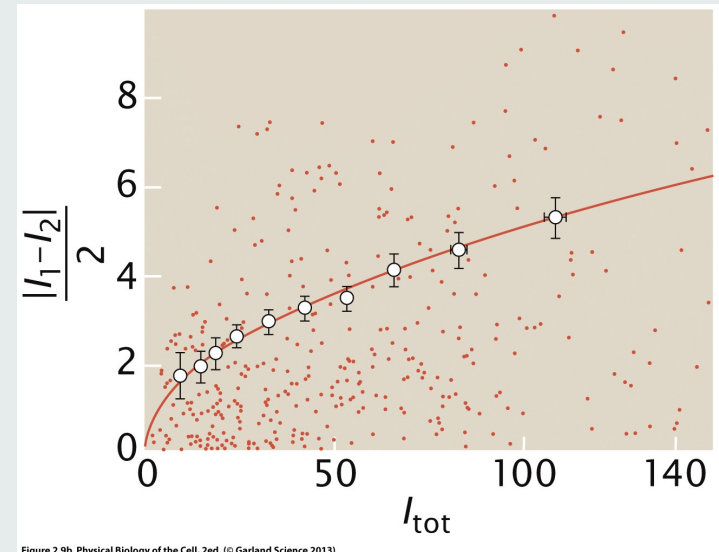


Figure 2.9b Physical Biology of the Cell, 2ed. (© Garland Science 2013)

Data from Rosenfield et al., *Science* 307:1962 (2005)

DO IT YOURSELF (*IN SILICO*)



1. Choose N (and α) and compute I_{tot}
2. Generate N random numbers to mimic cellular repartition (similar to what we did in the bootcamp...)
3. Compute the intensities of daughter cells I_1 and I_2
4. Repeat the “experiment” M times
5. Do the same for a different N

